

A de Polignac–Legendre formula for OEIS A129365: a proof of Bala’s Conjecture D

Adrian Perez Fontelles

Independent researcher

23 August 2026

Abstract

For OEIS A129365, $a(n) = (\prod_{j,k \leq n} \gcd(j, k)) / (\prod_{k=1}^n (\lfloor n/k \rfloor!)^k)$, P. Bala conjectured in April 2007 that, with $B = \text{A004125}$ the sum-of-remainders function, $\text{ord}_p a(np) = B(n) + B(\lfloor n/p \rfloor) + B(\lfloor n/p^2 \rfloor) + \cdots$ for every prime p and every $n \geq 1$ — a statement he described as reminiscent of de Polignac’s formula for $v_p(n!)$. We prove it. The route is the general closed form $\text{ord}_p a(m) = \sum_{i \geq 1} B(\lfloor m/p^i \rfloor)$, valid for every m ; Conjecture D is the specialisation $m = np$, where $\lfloor np/p^i \rfloor = \lfloor n/p^{i-1} \rfloor$ shifts the index by one and produces the leading term $B(n)$. The note is self-contained: the closed form is restated and reproved in Section 3.

1 The conjecture

OEIS A129365 (Peter Bala, 13 April 2007) has offset 1 and is defined by $a(n) = \text{A092287}(n)/\text{A129364}(n)$, equivalently, in the form recorded on the entry,

$$a(n) = \frac{\prod_{j=1}^n \prod_{k=1}^n \gcd(j, k)}{\prod_{k=1}^n (\lfloor n/k \rfloor!)^k}, \quad a(1), \dots, a(10) = 1, 1, 1, 1, 1, 2, 2, 2, 6, 48. \quad (1)$$

The entry lists four conjectures A–D. This note proves the last:

D) Let $b(n) = \text{A004125}(n)$. Then $\text{ord}_p(a(n \cdot p), p) = b(n) + b(\lfloor n/p \rfloor) + b(\lfloor n/p^2 \rfloor) + b(\lfloor n/p^3 \rfloor) + \cdots$. This is reminiscent of de Polignac’s formula (also due to Legendre) for the prime factorization of $n!$.

Conjectures B and C were addressed by T. Adamczewski (arXiv:2608.11941, August 2026), as annotated on the entry by M. De Vlieger on 20 August 2026. Conjecture A is proved in a companion note from the same closed form. As of the “Last modified” line on the live entry (23 August 2026), A and D are still recorded as unproven, and no proof of either appears in the entry’s links.

Throughout p is a prime, $v_p(N)$ is the exponent of p in N , and

$$B(M) = \sum_{k=1}^M (M \bmod k)$$

is A004125 (the entry’s b), with $B(0) = 0$.

2 Two elementary identities

Lemma 1 (nested floors). *For positive integers m, k, q , $\lfloor m/(kq) \rfloor = \lfloor \lfloor m/q \rfloor / k \rfloor$. In particular, for $q = p^i$ and $m = np$, $\lfloor np/p^i \rfloor = \lfloor n/p^{i-1} \rfloor$.*

Proof. For real x and a positive integer k one has $\lfloor x/k \rfloor = \lfloor \lfloor x \rfloor / k \rfloor$; apply this with $x = m/q$. The particular case is immediate from $np/p^i = n/p^{i-1}$. $\square$

Lemma 2 (remainder identity). *For every integer $M \geq 0$, $M^2 - \sum_{k=1}^M k \lfloor M/k \rfloor = B(M)$.*

Proof. $M^2 = \sum_{k=1}^M M$ and $M - k \lfloor M/k \rfloor = M \bmod k$ termwise. $\square$

3 The closed form

Theorem 3. *For every $m \geq 1$ and every prime p ,*

$$v_p(a(m)) = \sum_{i \geq 1} B(\lfloor m/p^i \rfloor).$$

Proof. Numerator. $v_p(\gcd(j, k)) = \min(v_p(j), v_p(k))$ is the number of $i \geq 1$ with $p^i \mid j$ and $p^i \mid k$. Summing over $j, k \leq m$ and exchanging the order of summation, the two conditions are independent, so

$$v_p\left(\prod_{j, k \leq m} \gcd(j, k)\right) = \sum_{i \geq 1} \lfloor m/p^i \rfloor^2.$$

(This is the square-case formula of A092287, confirmed on that entry by C. R. Greathouse IV in April 2013.)

Denominator. Put $M_i = \lfloor m/p^i \rfloor$. By Legendre's formula and Lemma 1,

$$v_p\left(\prod_{k=1}^m (\lfloor m/k \rfloor!)^k\right) = \sum_{k=1}^m k \sum_{i \geq 1} \lfloor \lfloor m/k \rfloor / p^i \rfloor = \sum_{i \geq 1} \sum_{k=1}^m k \lfloor m/(kp^i) \rfloor = \sum_{i \geq 1} \sum_{k=1}^{M_i} k \lfloor M_i/k \rfloor,$$

the last step again by Lemma 1 in the form $\lfloor m/(kp^i) \rfloor = \lfloor M_i/k \rfloor$, terms with $k > M_i$ vanishing.

Subtracting and applying Lemma 2 level by level gives $v_p(a(m)) = \sum_{i \geq 1} (M_i^2 - \sum_{k \leq M_i} k \lfloor M_i/k \rfloor) = \sum_{i \geq 1} B(M_i)$. $\square$

4 Conjecture D

Corollary 4 (Conjecture D). *For every $n \geq 1$ and every prime p ,*

$$\text{ord}_p a(np) = B(n) + B(\lfloor n/p \rfloor) + B(\lfloor n/p^2 \rfloor) + B(\lfloor n/p^3 \rfloor) + \cdots.$$

Proof. Apply Theorem 3 with $m = np$ and use the second part of Lemma 1: $\lfloor np/p^i \rfloor = \lfloor n/p^{i-1} \rfloor$ for every $i \geq 1$. Hence

$$v_p(a(np)) = \sum_{i \geq 1} B(\lfloor n/p^{i-1} \rfloor) = B(n) + B(\lfloor n/p \rfloor) + B(\lfloor n/p^2 \rfloor) + \cdots,$$

a finite sum since $B(0) = 0$. $\square$

Remark (the analogy Bala noticed is exact). Bala called D “reminiscent of de Polignac’s formula”. Theorem 3 shows the resemblance is not an analogy but the same shape: Legendre’s formula is $v_p(m!) = \sum_{i \geq 1} \lfloor m/p^i \rfloor$, i.e. the identity function evaluated along the sequence $\lfloor m/p^i \rfloor$, while here it is B evaluated along the same sequence. The shift to the block index $m = np$ is what puts the undivided $B(n)$ in front, and by the same computation $v_p(a(np + r))$ is the same for every $0 \leq r < p$, which is Conjecture C of the entry.

Remark (honest assessment). This note and its companion share Theorem 3; once the closed form is in hand, Conjecture D costs one substitution and Conjecture A costs one inequality. They are recorded separately because the entry states them as separate conjectures, not because the second required new work. The mathematics is elementary throughout and belongs in an OEIS comment.

A caution: a benchmark of 492 Lean-formalised open OEIS conjectures is currently being worked through by automated systems (Adamczewski, arXiv:2608.11941, 12 August 2026, reporting 147 resolved). Conjectures B and C of this very entry fell to that effort three days before this note was written. I could not confirm whether A or D are in that formalised set; check the results repositories before assuming priority.

5 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) recomputes $a(1), \dots, a(27)$ from (1) and matches the published DATA of A129365 exactly; (ii) prints $B(0), \dots, B(12) = 0, 0, 0, 1, 1, 4, 3, 8, 8, 12, 13, 22, 17$ and confirms $B(M) = 0 \iff M \leq 2$ for $M \leq 2000$; (iii) confirms Theorem 3 for every $1 \leq m \leq 60$ and every prime $p \leq m$, that is 597 pairs, with 0 mismatches; and (iv) confirms Corollary 4 directly, comparing $v_p(a(np))$ with $B(n) + B(\lfloor n/p \rfloor) + \dots$ for $p \in \{2, 3, 5, 7, 11, 13\}$ and all n with $np \leq 60$, that is 79 pairs, with 0 mismatches.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A129365>, entry A129365, consulted 23 August 2026; also entries A092287 and A004125.
- [2] A. M. Legendre, *Théorie des Nombres*, 3rd ed., Firmin Didot, Paris, 1830.
- [3] T. Adamczewski, *OEIS Open: How many conjectures can language models turn into theorems?*, arXiv:2608.11941 [cs.AI], 2026.

A Suggested OEIS comment (plain text, house style)

Conjecture D is true, and follows from a closed form valid for every index, not just multiples of p . With $B = A004125$, i.e. $B(M) = \text{Sum}_{\{k=1..M\}} (M \bmod k)$,

$$\text{ord}_p(a(m)) = \text{Sum}_{\{i \geq 1\}} B(\text{floor}(m/p^i)) \quad \text{for every } m \geq 1.$$

Proof: the numerator has $\text{ord}_p = \text{Sum}_i \text{floor}(m/p^i)^2$ (A092287); for the

denominator, Legendre together with $\text{floor}(m/(k \cdot p^i)) = \text{floor}(\text{floor}(m/p^i)/k)$ gives $\text{ord}_p = \sum_i \sum_{k=1..M_i} k \cdot \text{floor}(M_i/k)$, where $M_i = \text{floor}(m/p^i)$. Since $M^2 = \sum_{k=1..M} M$, the level- i difference is $\sum_{k=1..M_i} (M_i - k \cdot \text{floor}(M_i/k)) = \sum_{k=1..M_i} (M_i \bmod k) = B(M_i)$.

Conjecture D is the case $m = n \cdot p$: $\text{floor}(n \cdot p/p^i) = \text{floor}(n/p^{(i-1)})$, so the sum reindexes to $B(n) + B(\text{floor}(n/p)) + B(\text{floor}(n/p^2)) + \dots$. The same substitution with $m = n \cdot p + r$, $0 \leq r < p$, gives the same value, which is Conjecture C; and $B(M) \geq 0$ always, which is Conjecture A.